

Sensing

- **Review and Questions**
- **Lecture : Sensing**
- **Exercise**
- **Reflections**

Review

.....

Technē

Formal, codified, systematic knowledge oriented toward technical control and replicable production.



Mētis

Contextual, experiential, and embodied practical intelligence developed through lived engagement with particular situations and environments.

Development & Technology



What if development has *always* been about the application of technology – in its many forms – to the project of social transformation?

What if that's a problem?

Intervening in Complexity



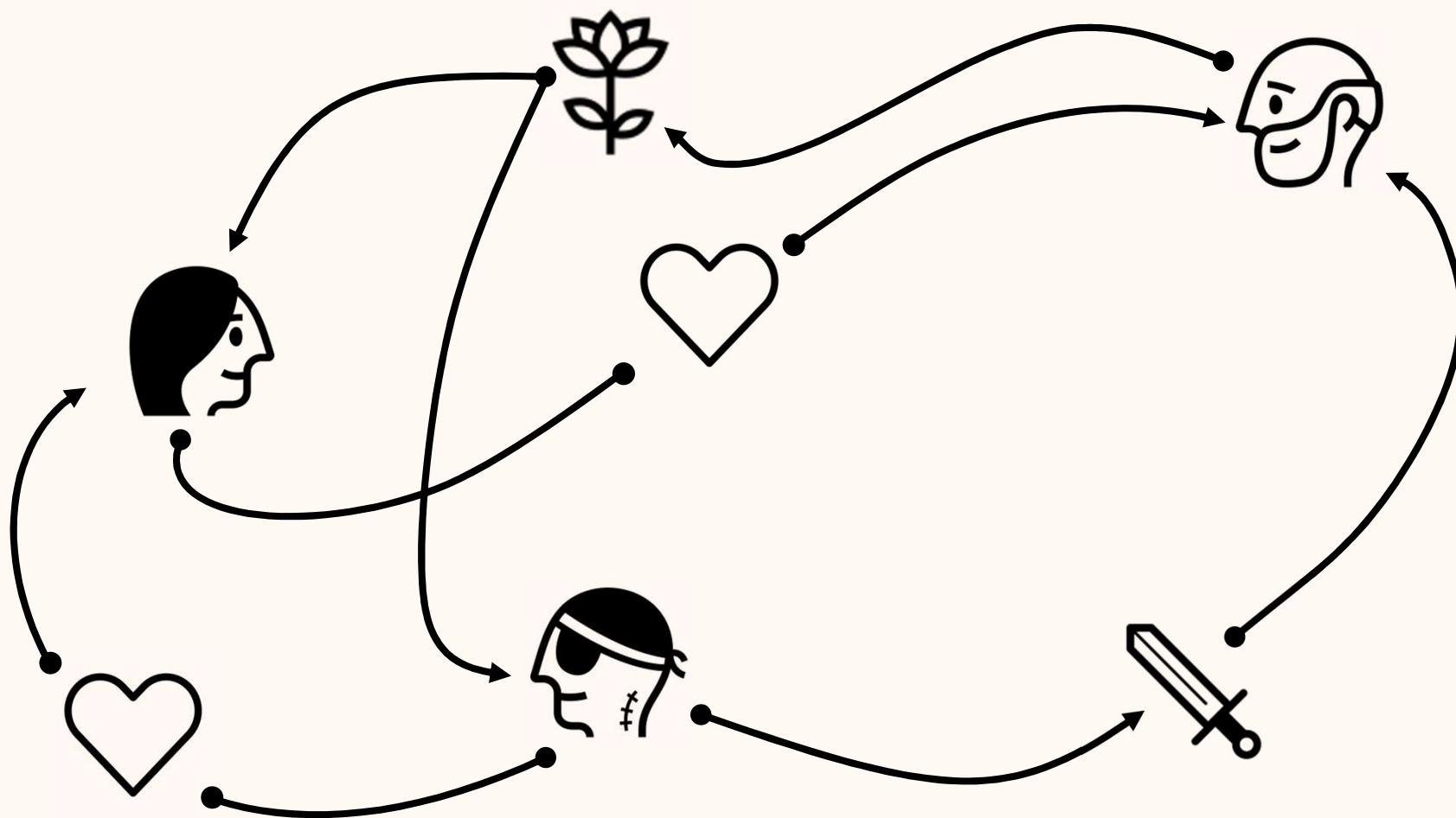
Snowden's Cynefin framework suggests the approach for discerning a path forward in complexity involves probing, sensing, and responding.



Design centers people...

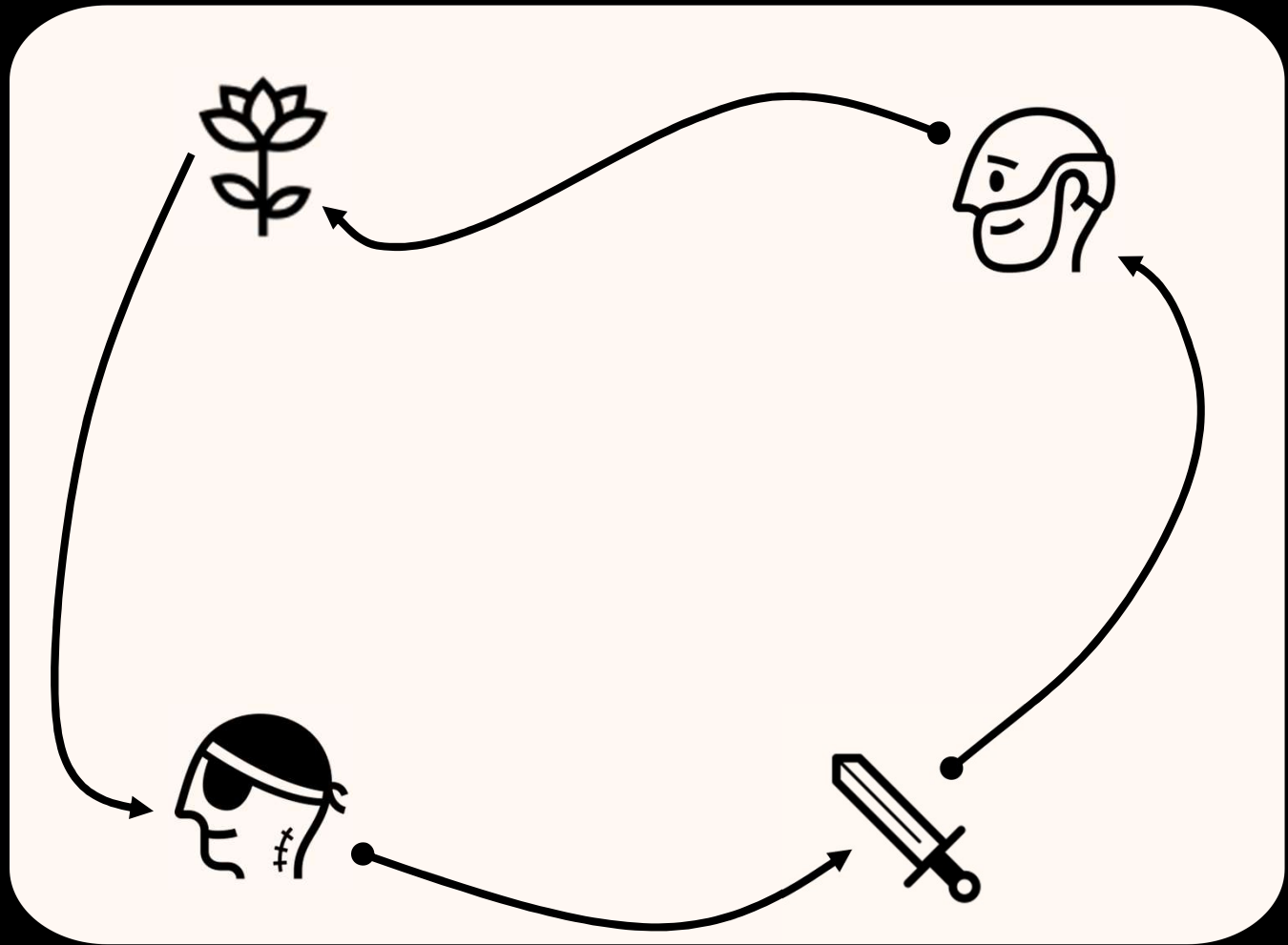


...but thinks in systems.



Design shifts frames.





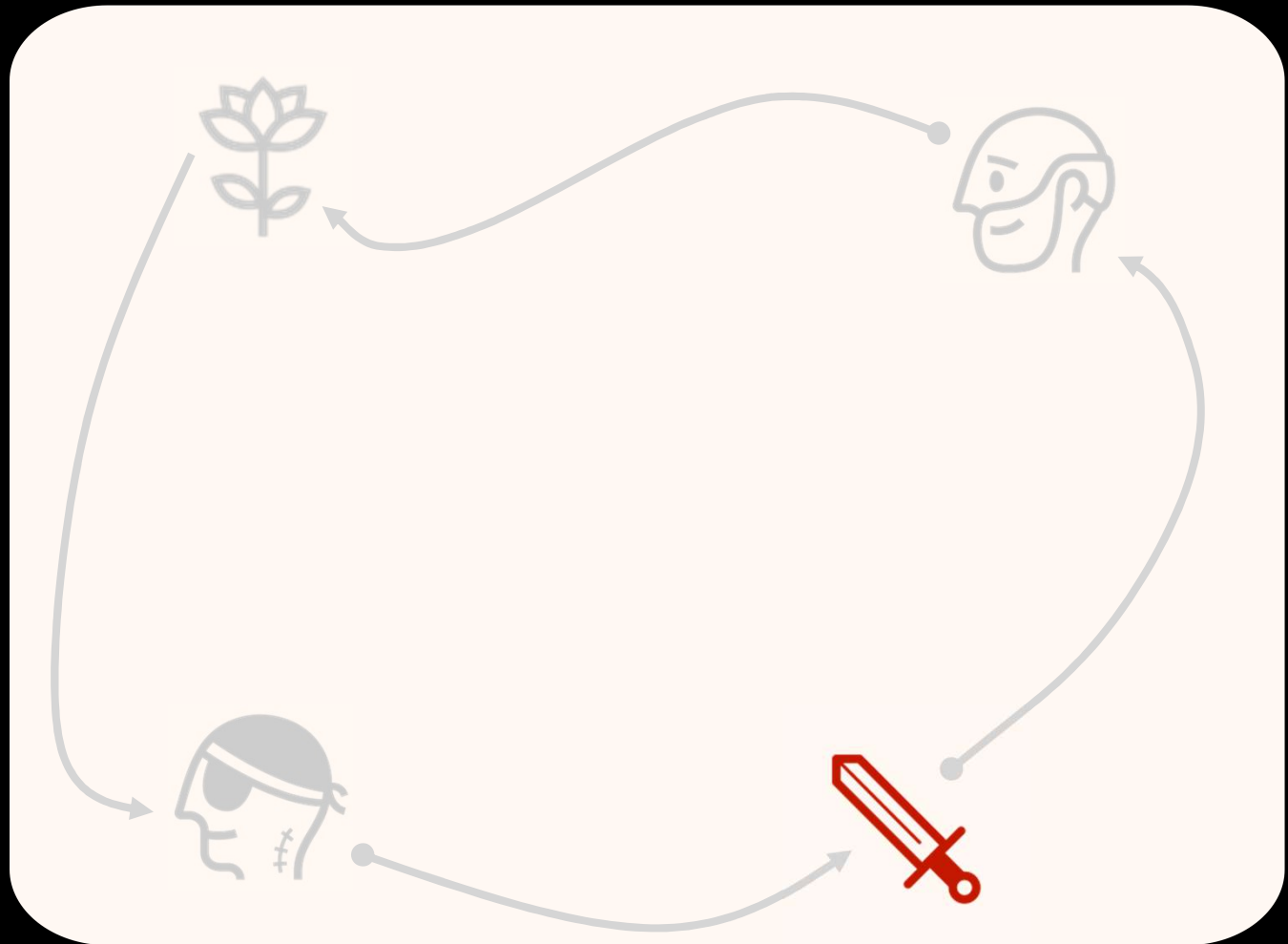


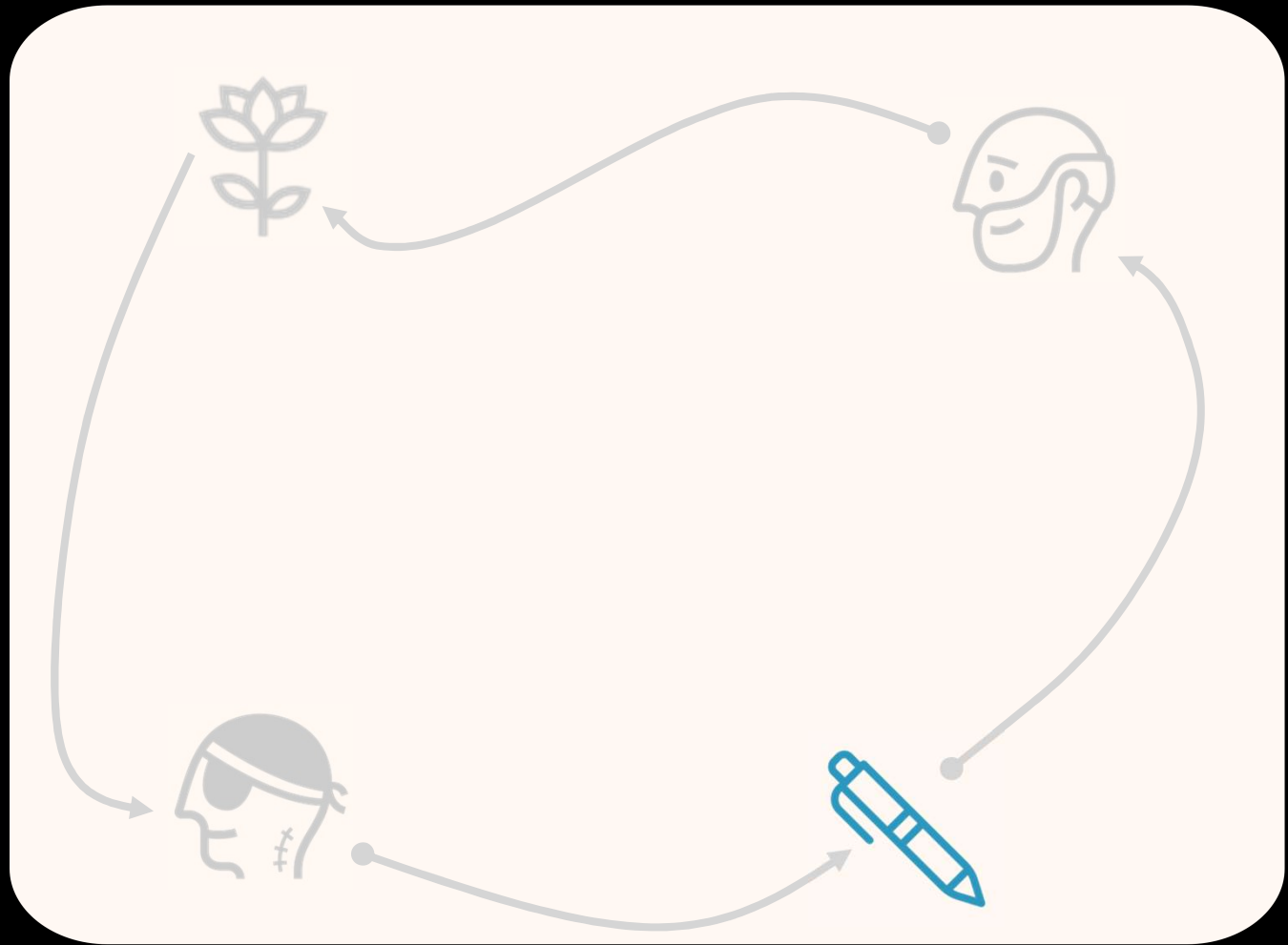
Design learns by trying.

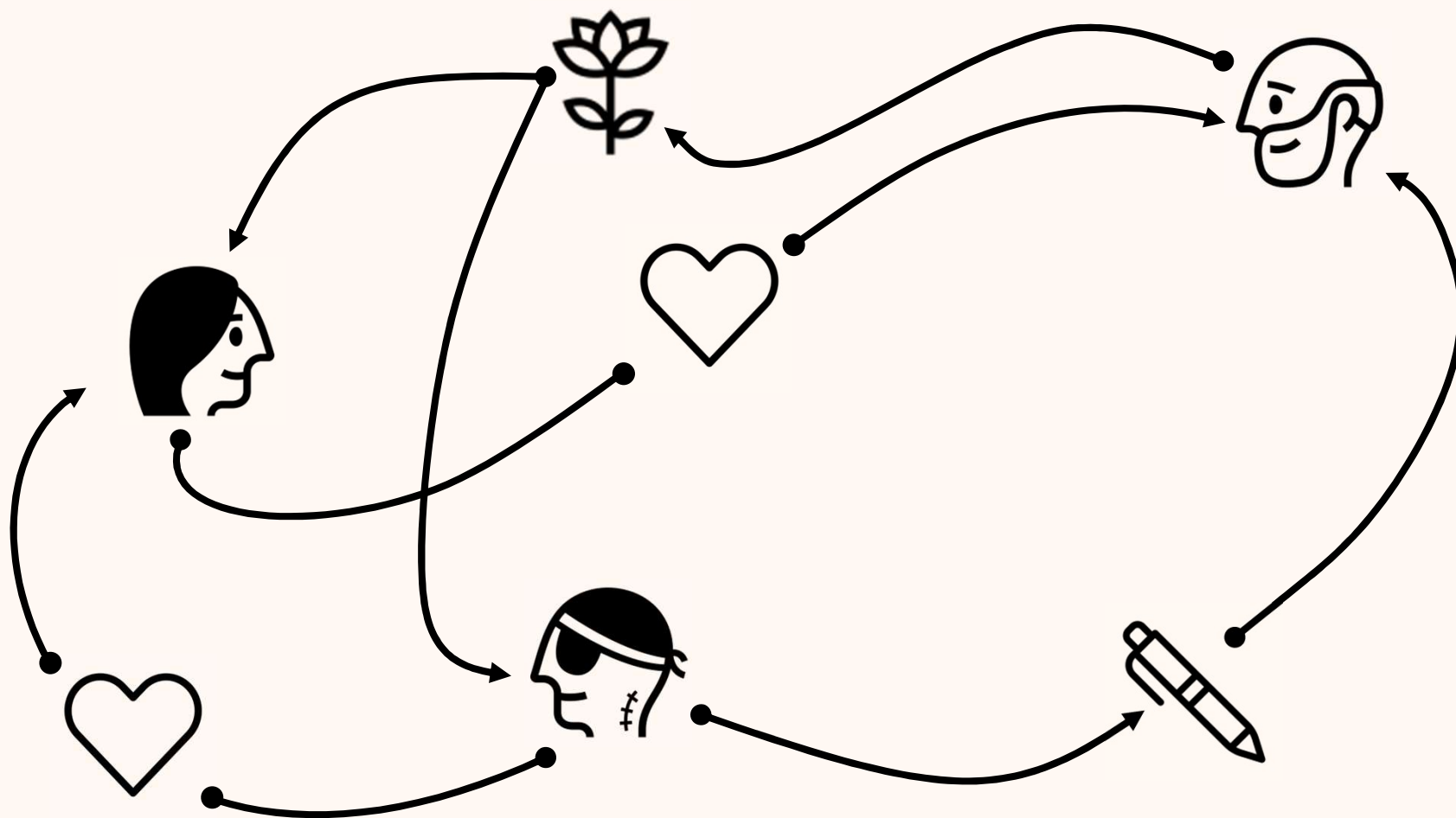
“

Through an active process of ideating, iterating, and critiquing potential solutions, design researchers continually reframe the problem as they attempt to make the right thing. The final output of this activity is a concrete problem framing and articulation of the preferred state, and a series of artifacts—models, prototypes, products, and documentation of the design process.”

Zimmerman, Forlizzi, and Evenson



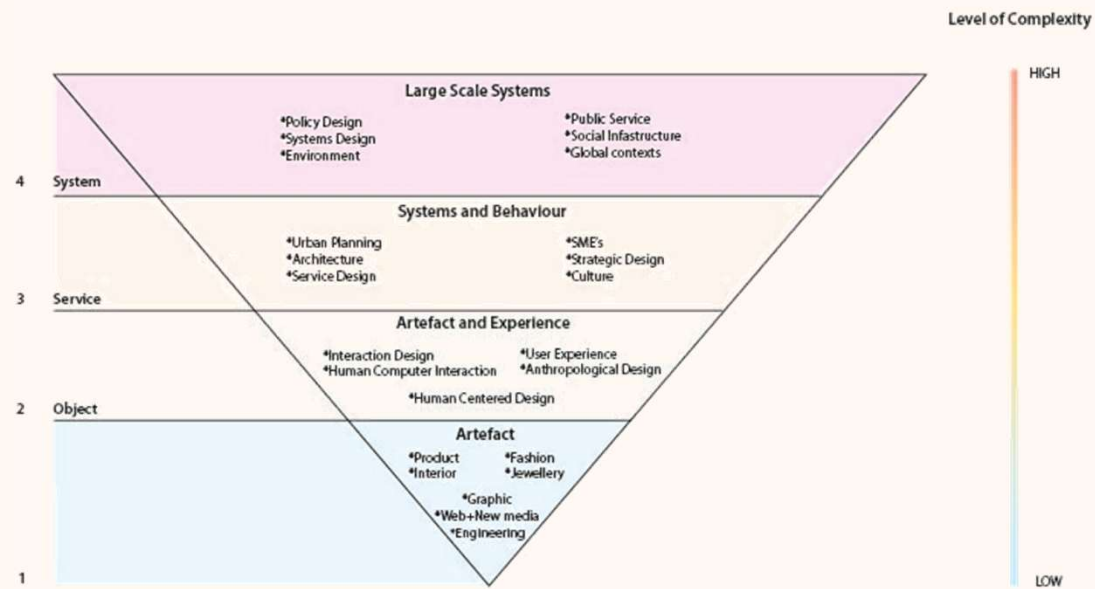




“

...composing an epic poem, executing a mural,
painting a masterpiece, writing a concerto.
But design is also cleaning and reorganizing a
desk drawer, pulling an impacted tooth,
baking an apple pie, choosing sides for a back-
lot baseball game, and educating a child.
Design is the conscious effort to impose
meaningful order.”

Papanek



Di Russo, 2014

Service Design is...



...the activity of choreographing people, infrastructure, communication, and material components of a service in order to create value for the multiple stakeholders involved.

(Mager)

Transition Design

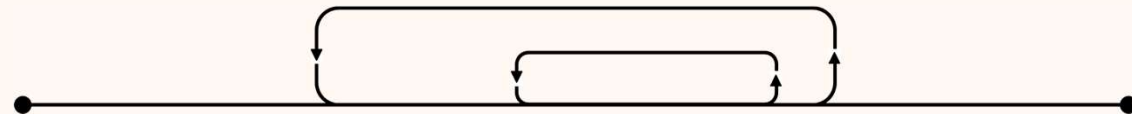


An approach for effecting long-term, transformation at scale of complex socio-technical-ecological systems.

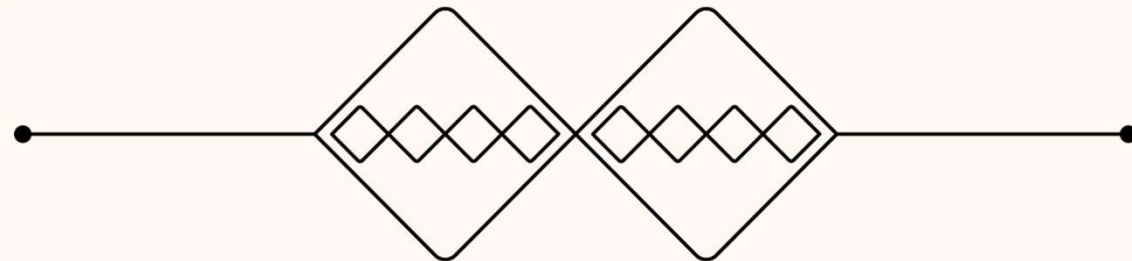
1. The Design Process

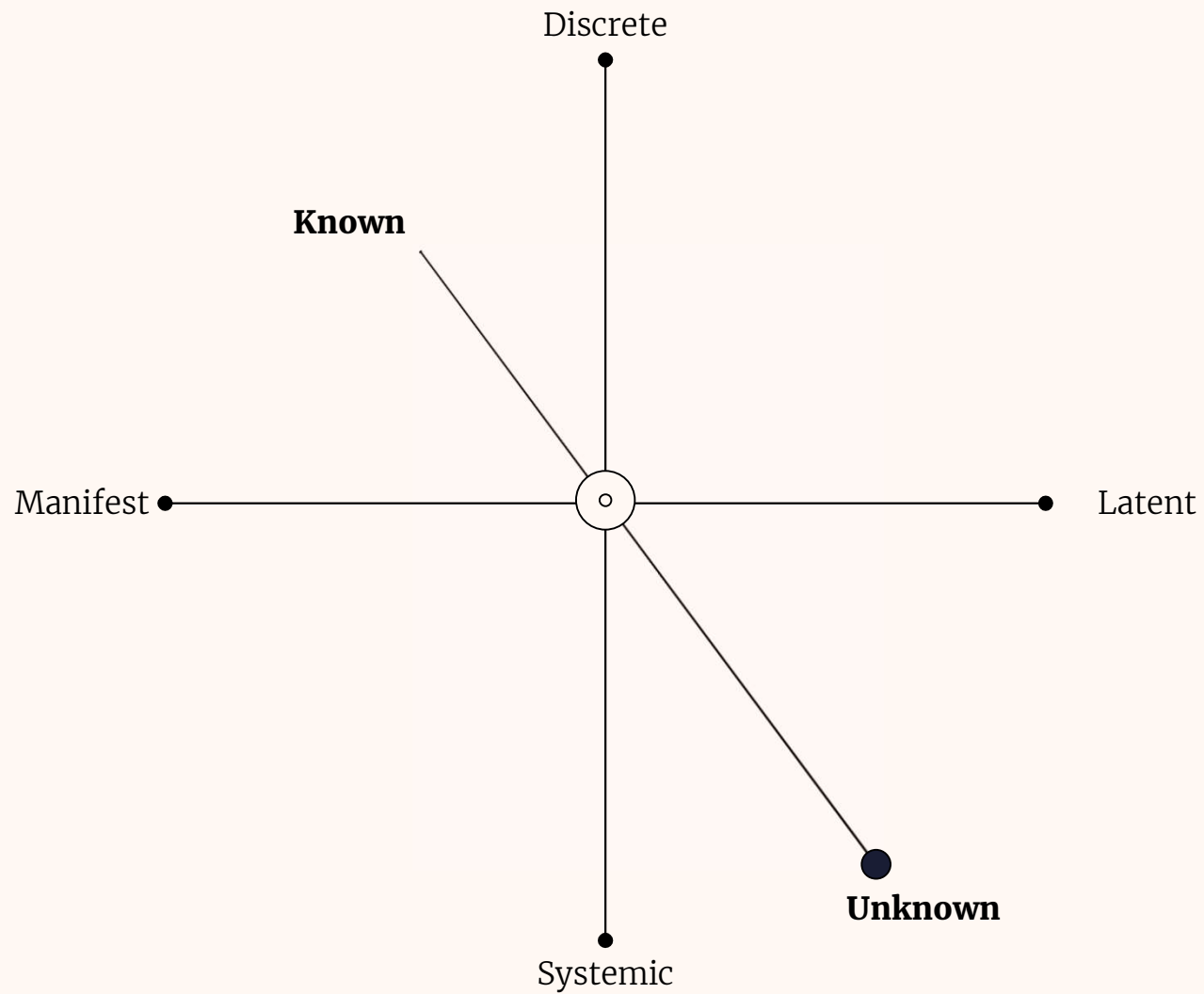


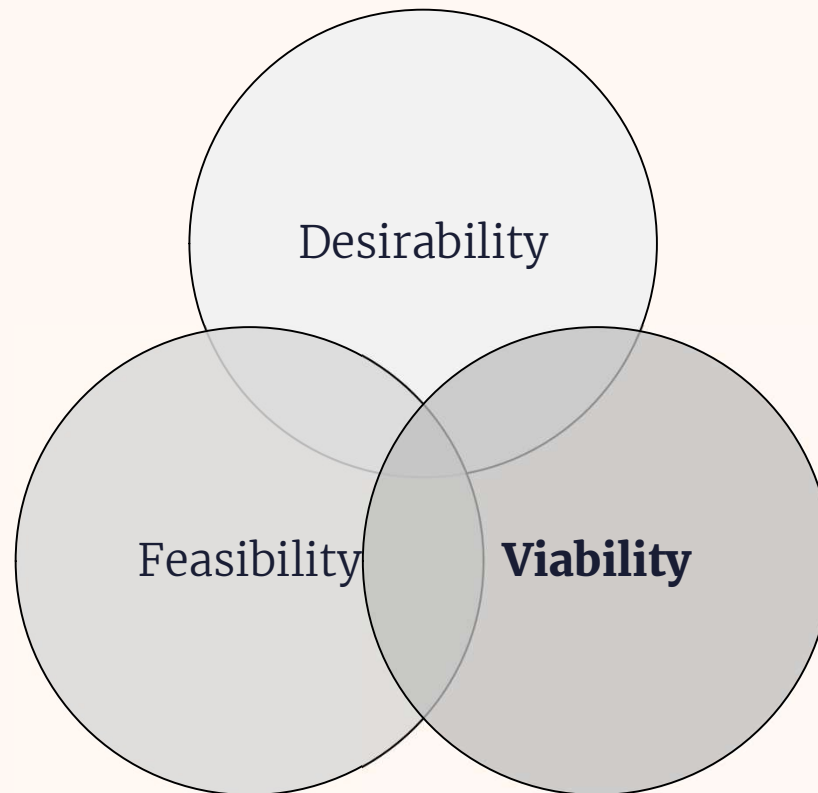
2. Iteration & Problem-Solution Coevolution



3. Divergence & Convergence

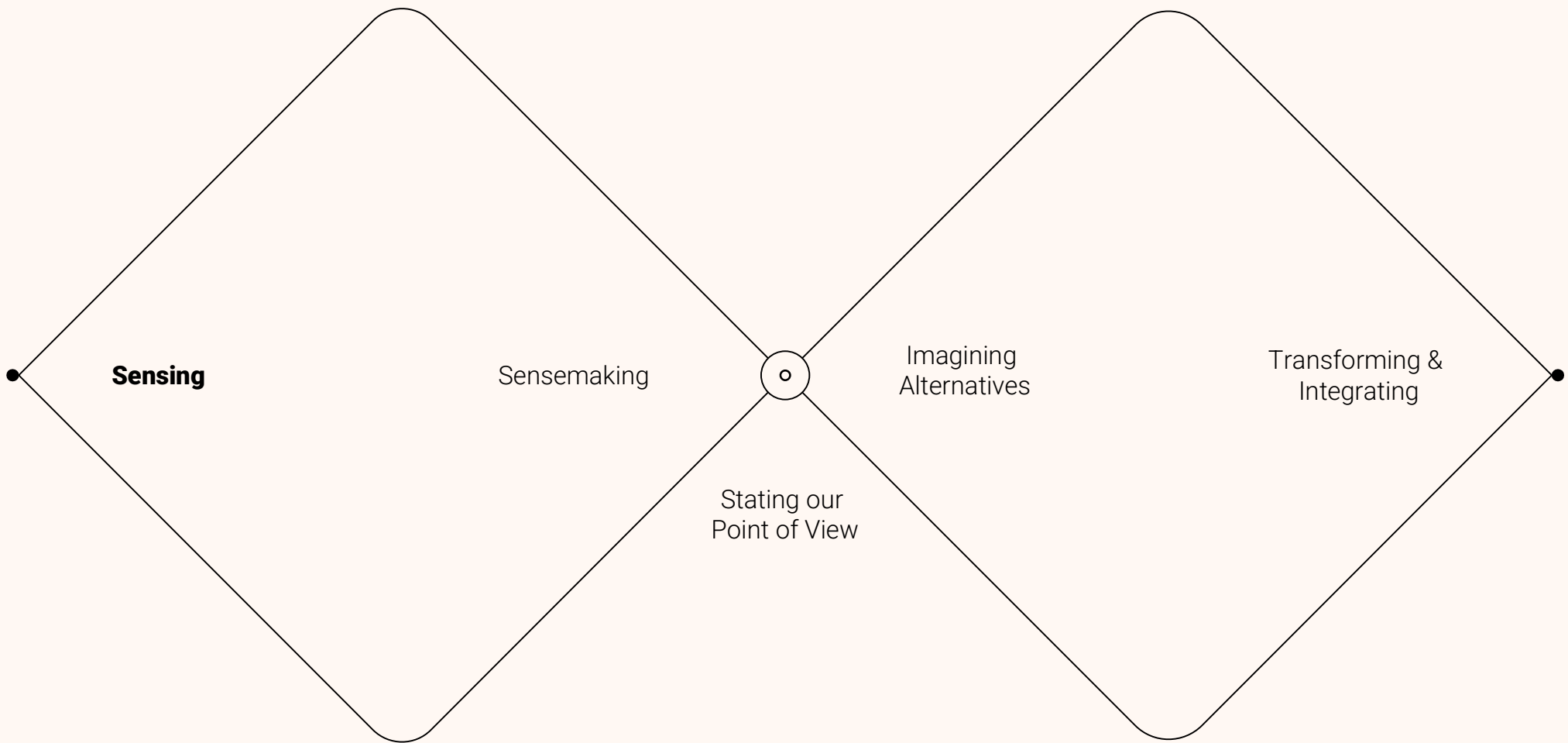






Sensing

.....



Sensing is...



...research.

Sensing is...



...research. Kind of.

Design Research



Design builds to learn.

Design Research



Design research shares elements with the scientific or *studious* research traditions, but acknowledges a creative goal.

Design Research

In the context of *doing* design, design research can be *research for design* or *research through design*.



Research *for* design is research that enables design.

Research *through* design is a “design activity that operates as research” (Forlizzi & Zimmerman).

Research *for* Design

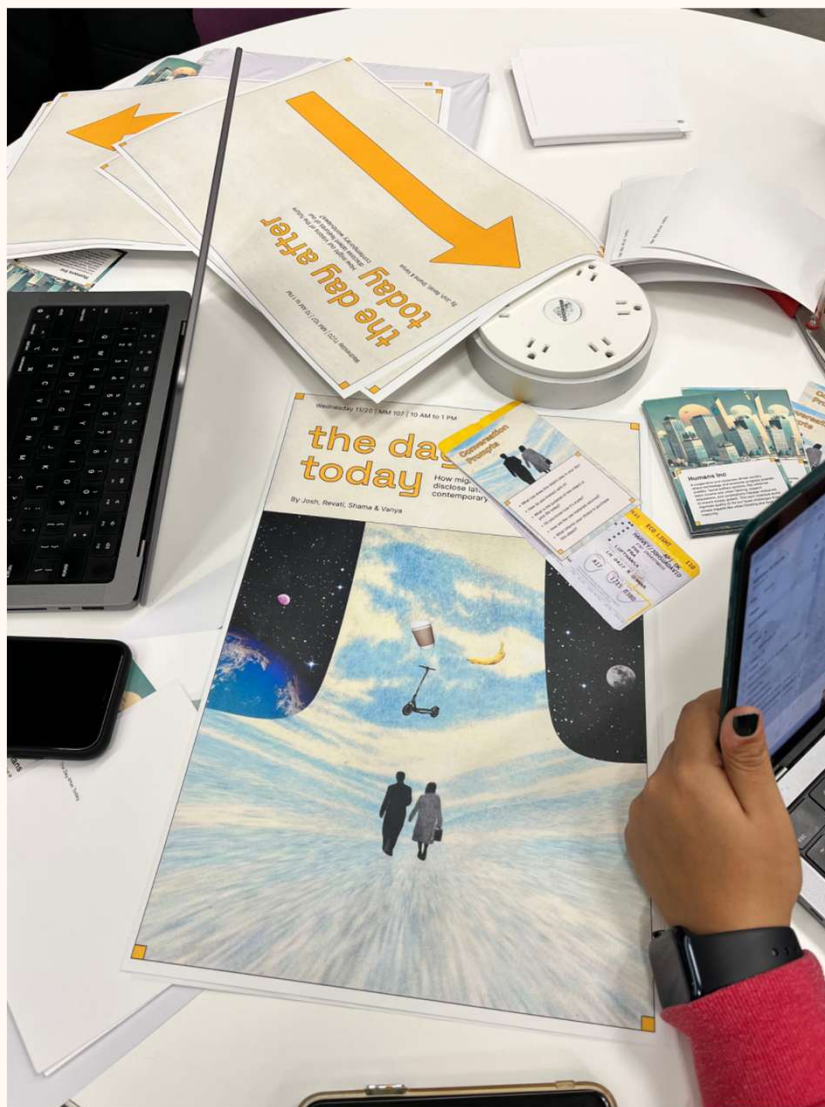


...research that enables design.

Research *through* Design



...is a “design activity that operates as research” (Forlizzi & Zimmerman).



Design Research



Surface and embed insights into/manifest insights as artefacts.

Artefacts are...



...instantiations of understanding that “reveal and become embodiments of possible futures”.



ABOUT LYNDIA Lyndia is from New Orleans – witnessing both the injustices and the resilience of her community during Katrina had a profound impact on her. She did the corps in New Orleans as a non-traditional Corps member; she's still in New Orleans but transitioned out of the classroom into school administration. She's passionate about education and is satisfied with the purpose in her work, but her present role isn't resonating with her personal goals.

Power to the people
cause the people have the power.

- Attends church and started social justice auxiliary.
- Is very community oriented and regularly attends community events; is well networked.
- Uses social to stay "plugged in".
- Mentors.
- Serves as an RST member with LEE.

- Hopes to support/impact community in a *big* way.
- Aspires to be a decision-maker to support community
- Values creating other leaders.

- Needs to know how to get access to decisionmakers.
- Wants career coaching.
- Wants an outlet for creating impact in civic life.
- Needs a support system.

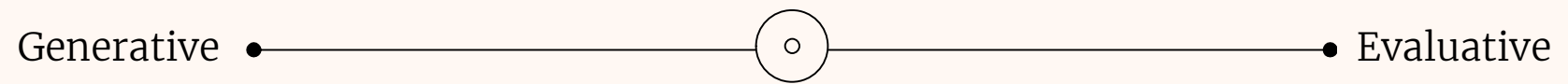
- Frustrated with complacency.
- Sees myriad injustices and worries that there's been little net improvement since she was a kid.

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Principles & Patterns



There are common principles, patterns, and purposes across research approaches.

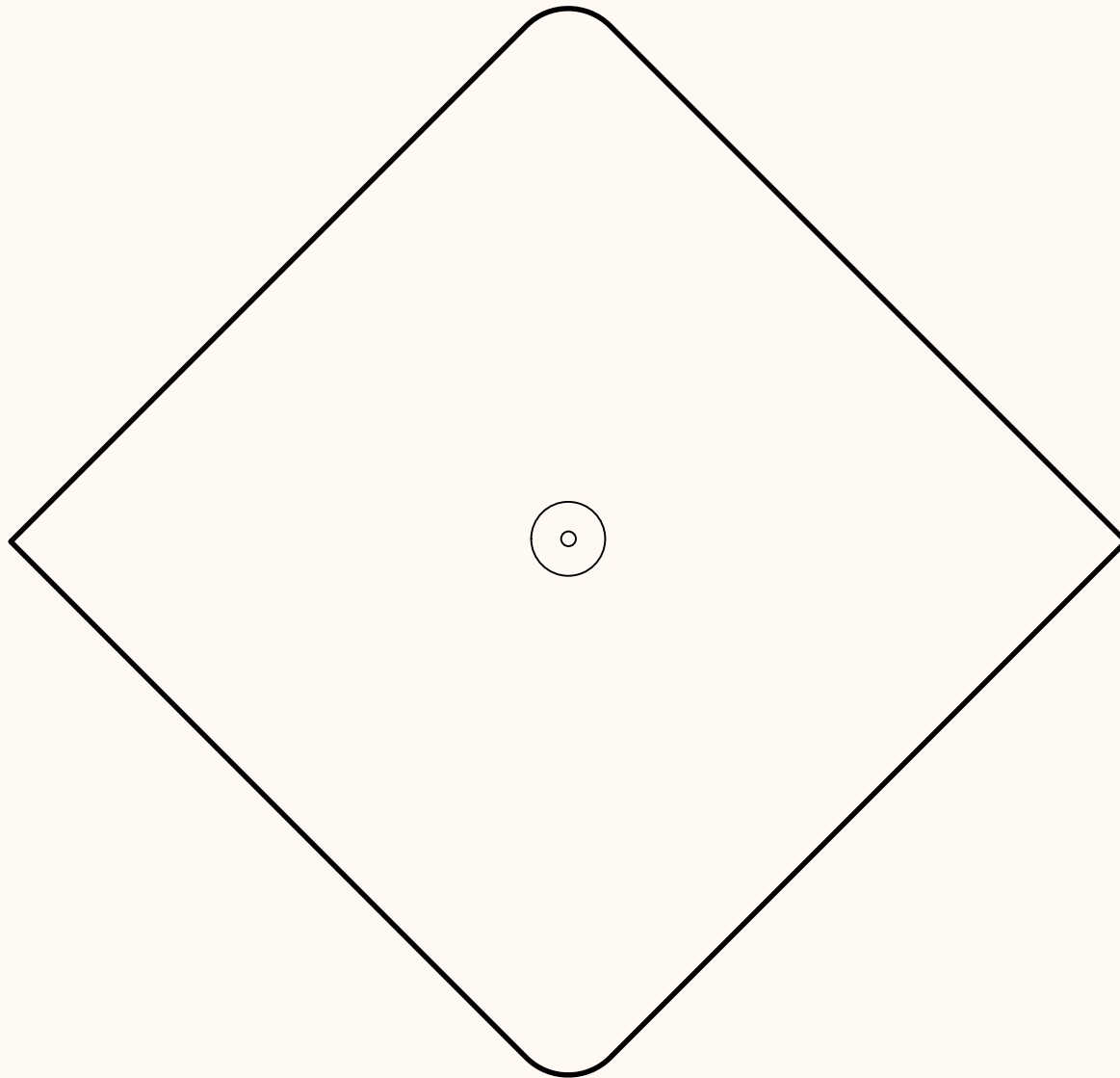


Remember

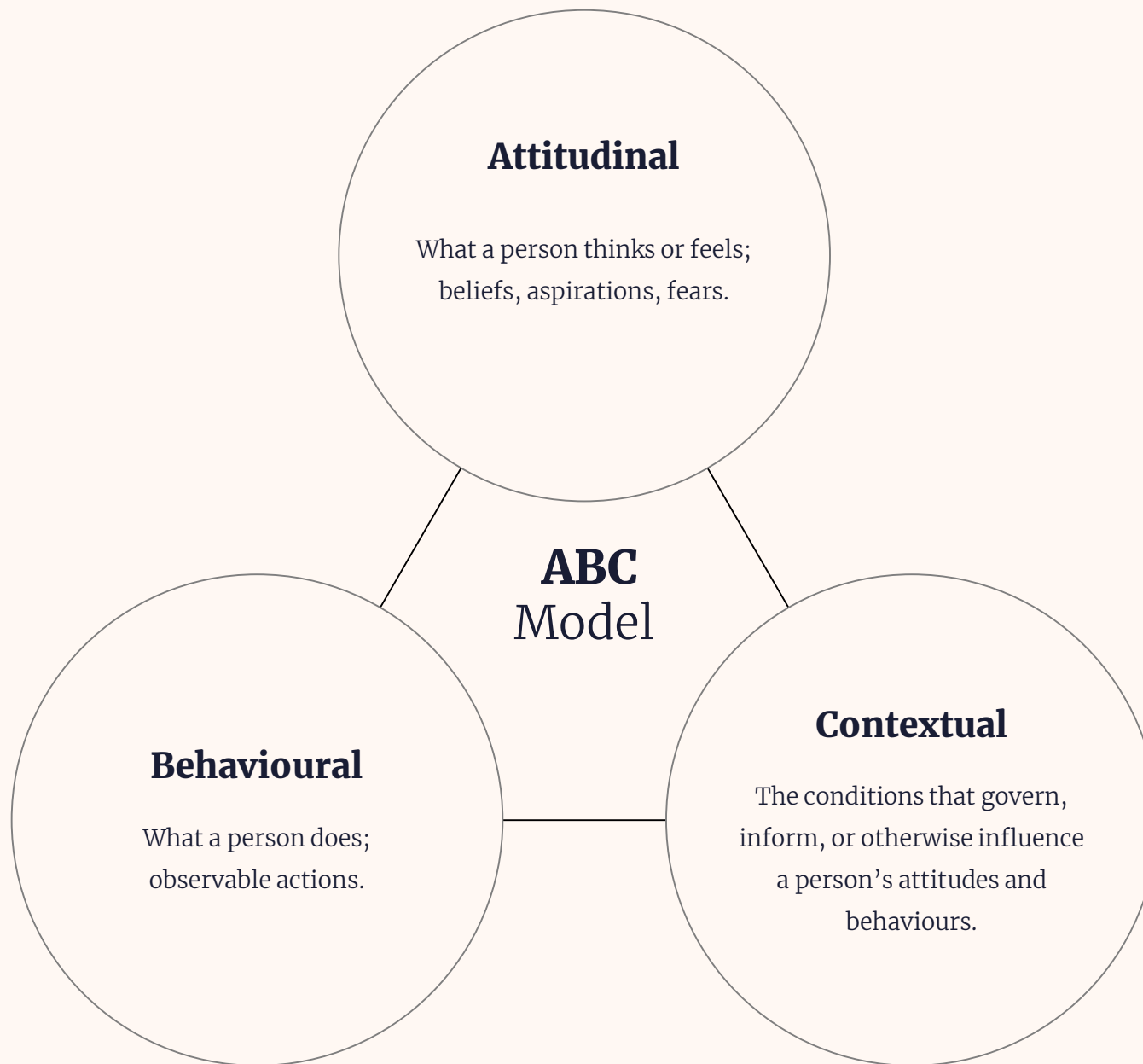


...divergence and convergence?

Generative



Evaluative



Remember



Design centers people, and thinks in systems.

Methods



Grounded in ethnographic methods, which explore cultural phenomena through the lens of people and relationships.

Ethnographic Methods

- Ethnography is learning from people: about their experience of the world, what they do and why they do it, what and how they think.

Ethnographic Methods



[Ethnography: Ellen Isaacs at TEDxBroadway](#)

Ethnographic Methods



Interview and observation are core.

Ethnographic interviews are...



- **Not a survey.**
- A dialogue with informants focus on what they know about their experience.
- Semi- or unstructured. Often in context. 20-60 minutes long.
- Made up of: explicit purpose, explanations, questions, repetition.
- **Not a survey.**

Descriptive

Sampling. “Can you tell me what you do?” “Could you describe an average day”

Structural

How informants understand. “What are all the thing you do to prepare for a lesson?” “What are all the types of vegetables you grow?”

Contrast

Meaning. “What’s the difference between a slide and an activity?” “What’s the difference between between a rake and a hoe?”

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Descriptive Questions



- **Grand Tour Questions**
 - **Typical** – “Tell me about a typical school day.”
 - **Specific** – “Tell me about school on Tuesday.”
 - **Guided** – “Show me around your school.”
 - **Task Related** – “Can I watch you prepare for your lesson and ask you questions afterward?”
- **Mini Tour Questions** – “Tell me about eating in the school cafeteria?”
- **Example Questions** – “You said Tuesday’s lunch was good. What are some examples of other good lunches?”
- **Experience Questions** – “What are some interesting experiences you’ve had in the school cafeteria?”
- **Native Language Questions** – “You said that lesson was a slog. What are some other things you might call a slog?”

Exercise



- **Think of a topic on which you would like to be interviewed. It must be something with which you have substantial personal experience: a job, a hobby (skiing, gaming, knitting), an activity in which you regularly engage (voting, volunteering).**
- **Exchange topics with your partner. You will each have 5 minutes to prepare to interview your partner.**
- **Take turns interviewing one another for approximately 20 minutes each.**
- **Try the various question types with emphasis on the several categories of descriptive question.**

Reflections

- Shift power: asymmetry and informant direction.
- Let patterns emerge; don't force it.
- ABCs – Always Be Capturing
- Drive toward *meaning*, as opposed to narrative.
- Move from large to small to large.

Additional Methods



Grounded in interview and observation.

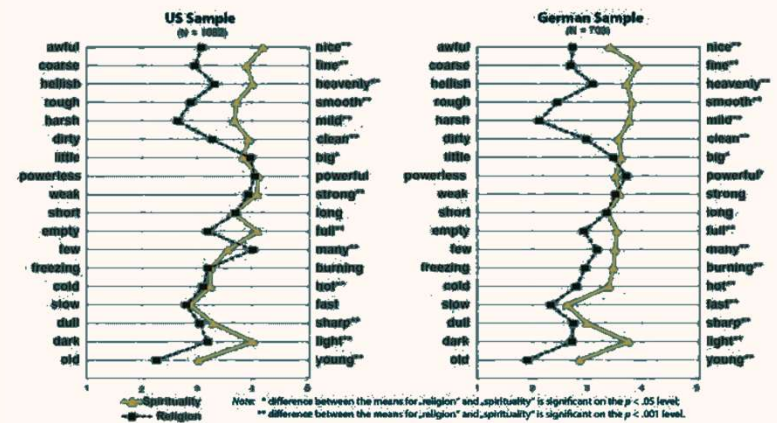
Card Sort

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Semantic Differential

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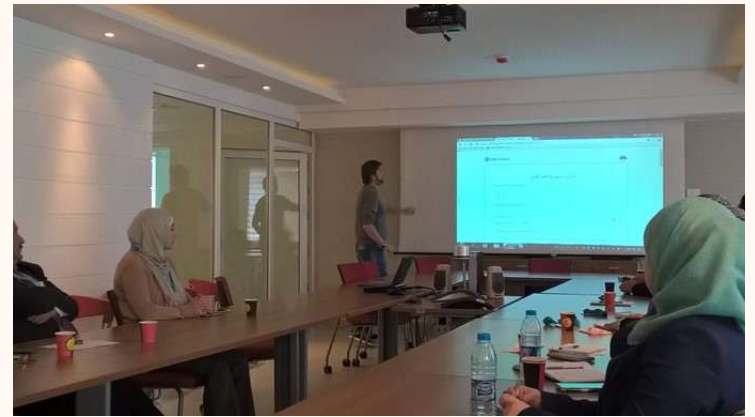
Diad/Triad Interviews



Contextual Inquiry



Focus Groups



Focus groups



Diary Study



Cultural Probe

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•
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Capturing

- Always Be Capturing
- Interviews are primary, observations are contextual
- Verbatim Principle
- Maintain awareness of bias/interpretation
- Capture necessary details – location, time, physical artifacts (Place), demographic and contact information (Actors), and Activities.

Sensemaking

What are we going to do with all this information?



Sensemaking is...



“

**...structuring the unknown so as to be able to
act on it.”**

Waterman

Sensemaking

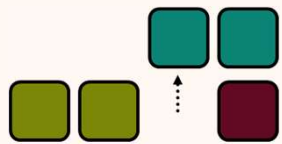


It is the process of “structuring the unknown” (Waterman, 1990, p. 41) by “placing stimuli into some kind of framework” that enables us “to comprehend, understand, explain, attribute, extrapolate, and predict” (Starbuck & Milliken, 1988, p. 51).

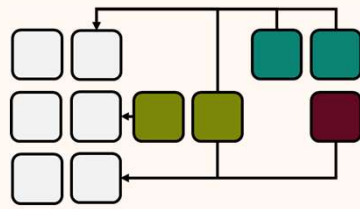
Sensemaking



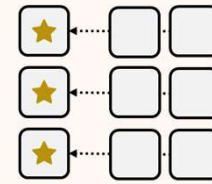
Involves four activities:



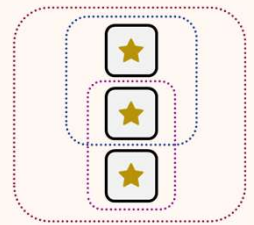
Organize Findings



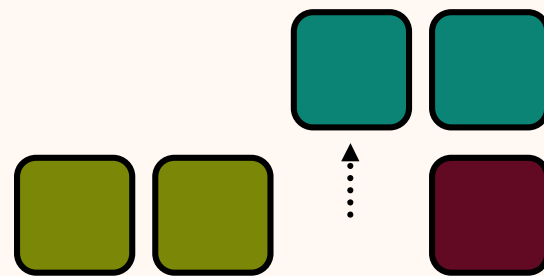
Find Themes



Identify Insights



Create Frames



Organize Findings

Sensemaking



One way to organize is by affinity.

Affinity Diagraming

⋮

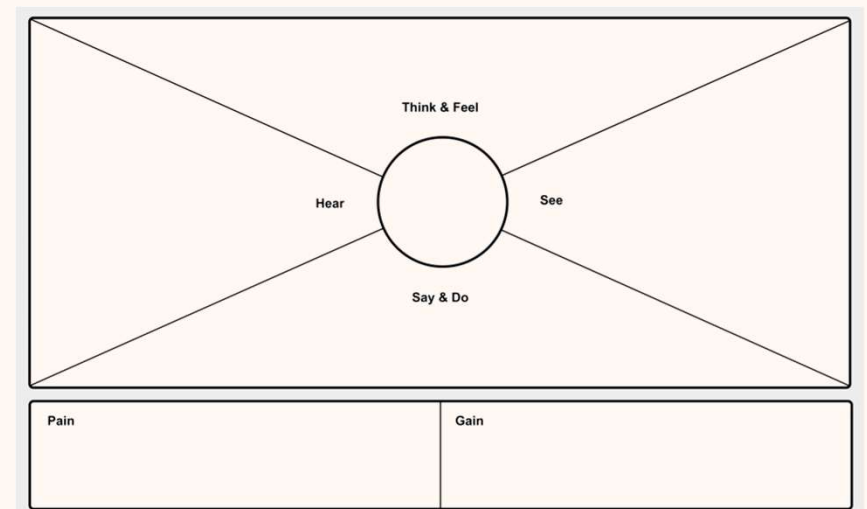


Sensemaking



Another is through the creation of artefacts:

Empathy Maps



Personas



LYNDA JACKSON

1.0 | Hypothetical



AGE
32

GENDER/IDENTITY
Woman

RACE/ETHNICITY
New Orleans

LOCATION
District Administrator

OCCUPATION
M.A. Education Policy

MARITAL/PARTNER STATUS
Single

FAMILY
Lives alone

HOMETOWN HERO

ABOUT LYNDA Lynda is from New Orleans – witnessing both the injustices and the resilience of her community during Katrina had a profound impact on her. She did the corps in New Orleans as a non-traditional Corps member; she's still in New Orleans but transitioned out of the classroom into school administration. She's passionate about education and is satisfied with the purpose in her work, but her present role isn't resonating with her personal goals.

LYNDA SAYS

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BEHAVIOURS

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ASPIRATIONS
+ GOALS

- Hopes to support/impact community in a *big* way.
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- Values creating other leaders.

WANTS
+ NEEDS

- Needs to know how to get access to decisionmakers.
- Wants career coaching.
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FRUSTRATIONS
+ PAIN POINTS

- Frustrated with complacency.
- Sees myriad injustices and worries that there's been little net improvement since she was a kid.

Unit 2 Module 1.2 • Lecture

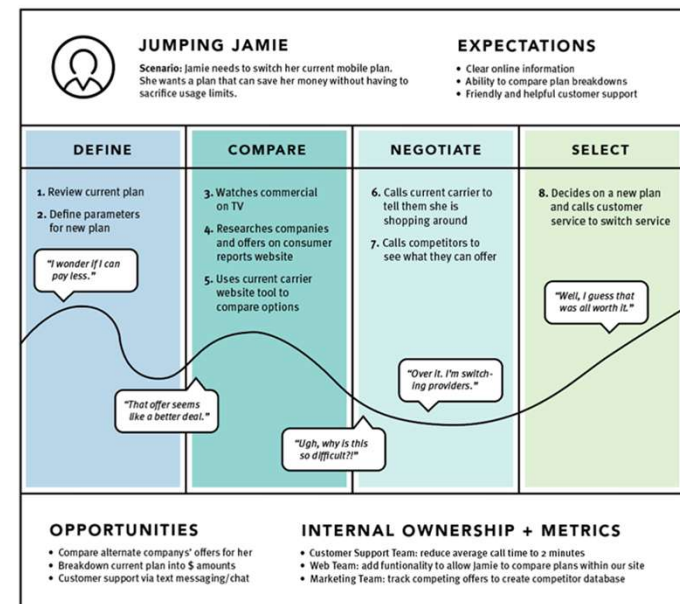
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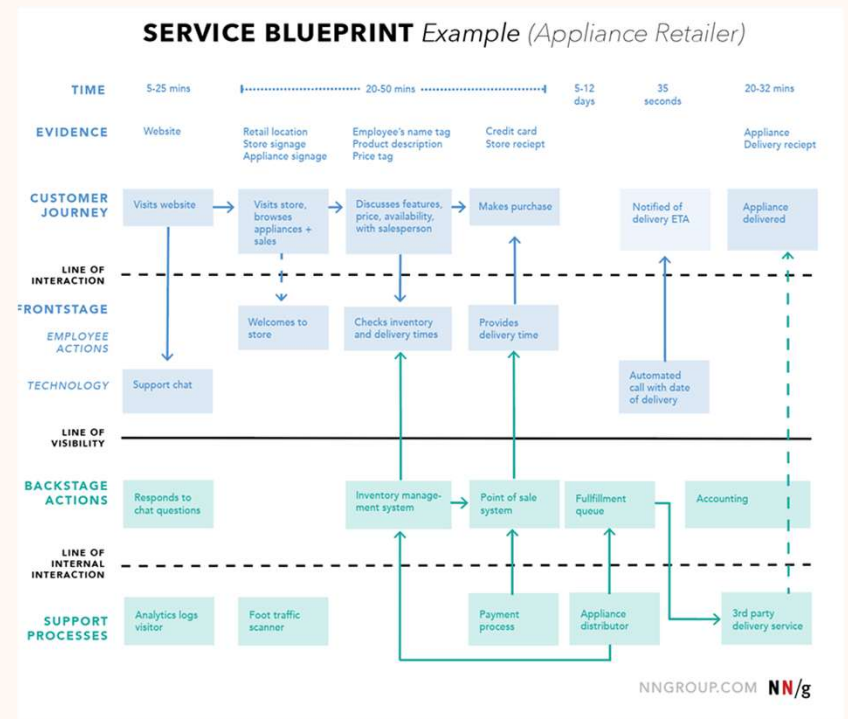
Journey Maps



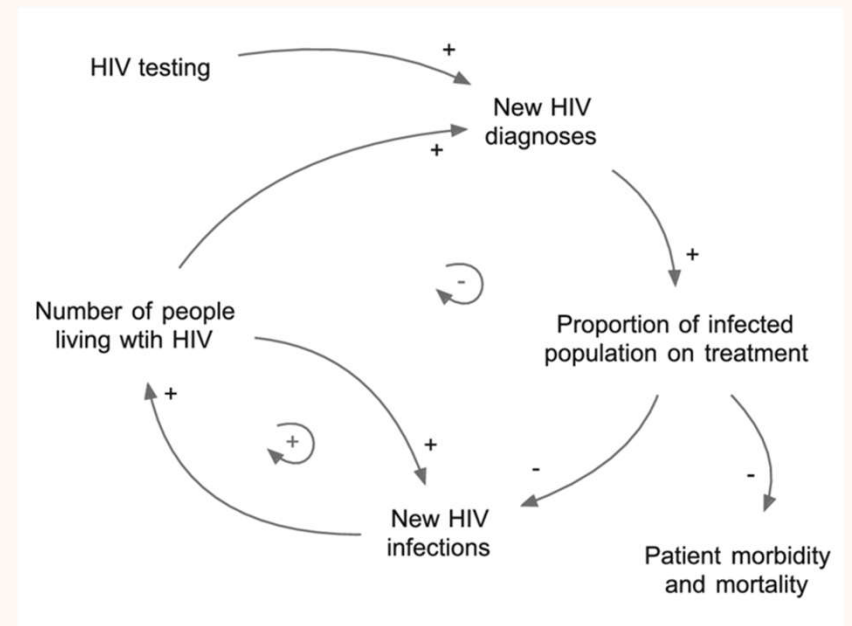
CUSTOMER JOURNEY MAP Example (Switching Mobile Plans)



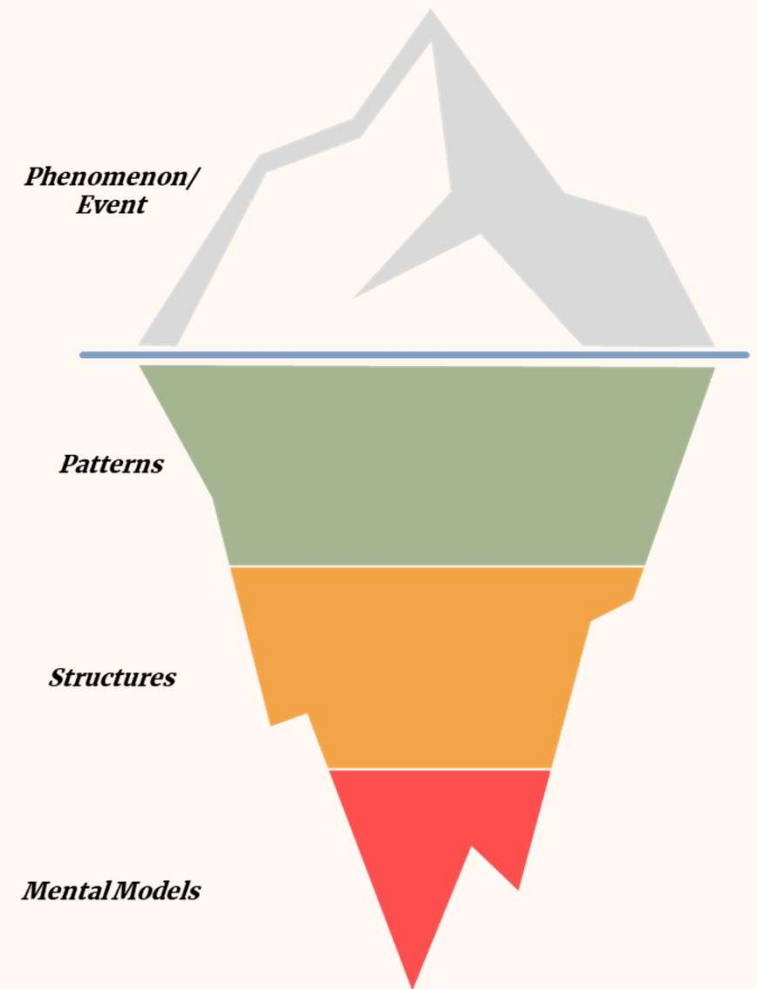
Service Blueprints

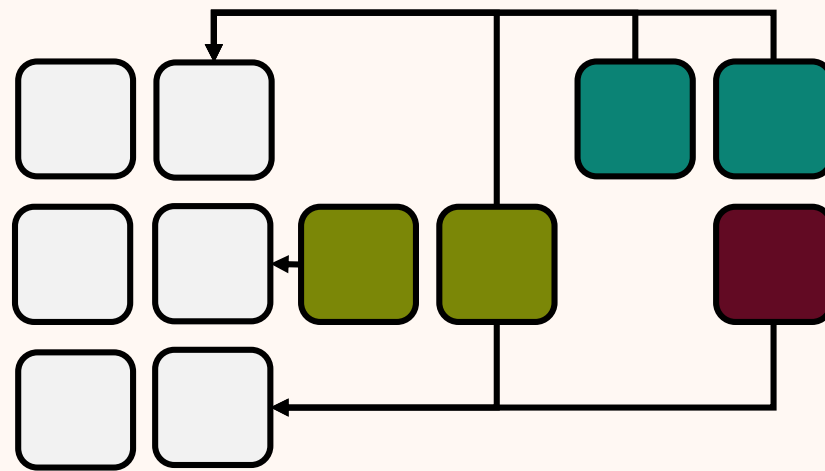


Causal Loop Diagrams



Iceberg Diagrams





Find Themes

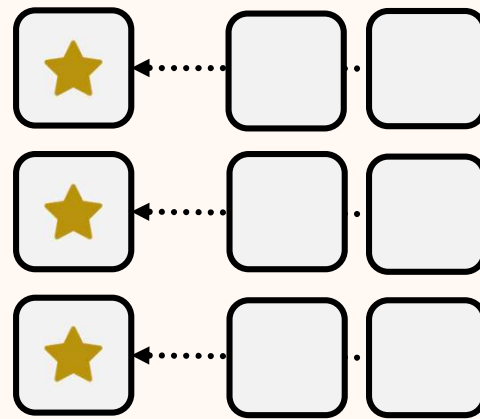
Themes

- Themes are patterns, often relating to the behaviours and experiences of actors and the needs and motivations expressed through those behaviours.

Coding



1	Interview 5 – Patient 5
2	
3	MC: Can you tell me how you feel about your experience of intensive care?
4	
5	Patient 5: Yes. I was admitted to hospital with a chest infection. It just got
6	worse and worse and I was struggling to breathe. I remember the doctor
7	coming to see me and I could tell she thought I was unwell. She stabbed me
8	in the wrist with a needle and then when she came back there seemed to be a
9	bit of a panic. I remember her explaining to me that I might need to go to
10	intensive care and I may end up on a ventilator, which I found really scary.
11	
12	MC: What did you find scary?
13	
14	Patient 5: It was that she would put me to sleep and I might not wake up.
15	
16	Red = Reason for admission
17	Yellow = Referral to ICU
18	Green = Patients perception of staff
19	Turquoise = Painful procedure
20	Blue = Treatment plan for admission and escalation
21	Pink = Patient expressing anxieties
22	



Identify Insights

1

Well-informed.

Is it informed by multiple data points—including secondary research, lived experience, and subject matter expertise?

2

More than an observation.

Does it offer insight into how or why a phenomenon is occurring?
Does it offer a compelling reframe of something we already know?

3

So what?

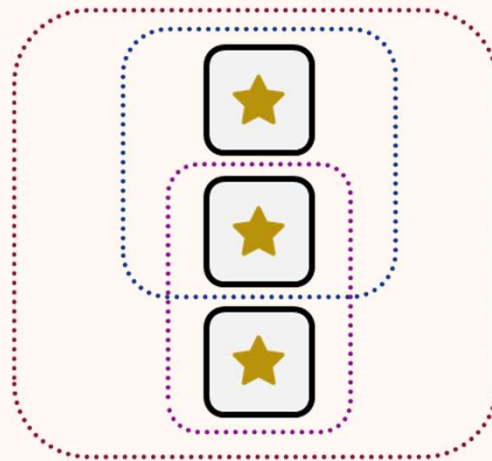
Does it help people understand why it matters? Does it capture a tension or a shift that needs to happen? Does it connect to project impact objectives and learning aims?

Design Opportunities

Generative insights

Design Principles/ Constraints

Evaluative insights.



Create Frames

Frames



Frames are thinking tools that act as lenses for explaining and interpreting the problem space.

Frames



Framing is about reflecting the facts of a problem, but doing so in a way that affords opportunity.

Frames

Frames follow a format:

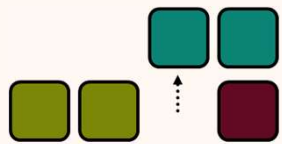


If the problem situation is approached as if it is...,
then....

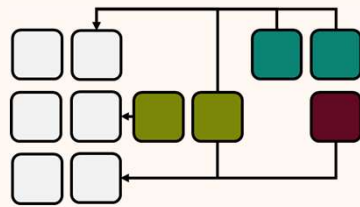
Exercise



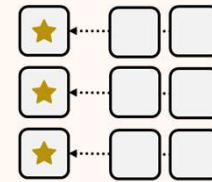
- **Head to** [Frame Creation and Design in the Expanded Field - ScienceDirect](#)
- **Read the case study** (*Italicized and beginning: "A brief example should help to explain the workings of this frame creation process:"*)
- **Discuss.** What are your thoughts on this approach? What does it make possible? What are its strengths? Shortcoming?



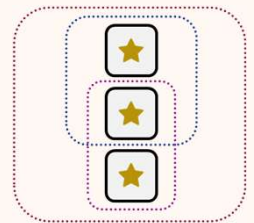
Organize Findings



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